The University of Sydney

Master of Professional Engineering (Electrical)

ELEC5020: Capstone Project A

Project Proposal

"*Segmentation of Vocal Tract using real time MRI data*"

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Project Link: https://github.com/Tusharbhai07/Segmentation-of-Vocal-Tract-using-rt-MRI.git

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Declaration

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Sincerely,

Tusharkhupte

Tushar Manish Khupte

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1. Project Overview

This project focuses on segmenting and analysing the vocal tract using rt-MRI images, with the help of modern AI models like the Segment Anything Model (SAM3). Unlike traditional speech analysis which only uses audio signals MRI allows direct visualization of articulatory motion during speech production. These include structures such as the tongue, lips, jaw, palate and airway which are all important for producing speech sounds.

However, accurately identifying and separating these structures in MRI images is still difficult. This is because MRI data can be noisy, varies between people and involves continuous movement during speech. While recent methods like SAM-based models have improved segmentation most of them only focus on generating visual masks and do not go further into analysing speech-related information.

The main goal of this project is to build an automated system that can both segment and analyse vocal tract MRI data. First, SAM3 will be used to generate segmentation masks of key speech organs from MRI frames. Then, these segmented regions will be used to extract useful articulatory variables such as lip aperture (LA), tongue position (TT, TB), velum position (VL), and airway geometry. These features will also be studied over time to understand how they change during speech.

From a technical point of view, the system combines computer vision and signal processing techniques. SAM3 helps in accurately identifying different regions with minimal training and additional processing is used to measure shapes and areas from the segmented images. By analysing multiple frames the system can capture how speech organs move dynamically.

This system can be useful in several real-world applications:

- In speech research it helps in understanding how different sounds are physically produced.
- In healthcare it can support the diagnosis and monitoring of speech disorders.
- In speech synthesis it can improve how artificial voices are generated by making them more realistic.
- In AI applications it can improve systems that combine both audio and visual information for better speech understanding.

Overall, this project goes beyond just segmentation by using AI to extract meaningful information from MRI data. It connects computer vision with speech science making it useful

2. Background and Motivation

Vocal tract MRI analysis has become an important research area in speech science, articulatory modelling and clinical speech assessment as it provides direct visualization of internal speech organ movements. Unlike traditional acoustic-based methods real-time MRI (rt-MRI) enables detailed observation of articulators such as the tongue, lips, jaw, palate, and airway offering deeper insight into the physical mechanisms of speech production. However, analysing vocal tract MRI data remains challenging due to low image contrast, noise, complex articulator motion and the large volume of temporal data.

Early work by Bresch and Narayanan (2009) introduced frequency-domain segmentation techniques for upper-airway MRI. While this approach was influential, it relied on anatomically informed modelling and frame-wise processing making it difficult to scale for continuous speech analysis and limiting full automation[2].

Subsequent research shifted toward deep learning-based segmentation. For example, Ruthven et al. (2021) demonstrated that convolutional neural networks (CNNs) particularly U-Net architectures can significantly improve segmentation accuracy of vocal tract structures. However, these approaches depend heavily on large annotated datasets and often struggle to generalize across different speakers and imaging conditions[1].

Silva and Teixeira (2015) proposed an unsupervised framework for vocal tract segmentation from real-time MRI, leveraging a modified Active Appearance Model to capture both shape and intensity variations without requiring labelled training data. While this approach significantly improved automation and reduced reliance on manual annotation, its performance still depends on model initialization and may struggle with large inter-speaker variability and complex articulatory dynamics, indicating challenges in achieving fully robust and generalizable segmentation [3]

More recently, the introduction of foundation models has transformed the field of image segmentation. Hsu et al. (2026) applied SAM 2.0 to real-time vocal tract MRI and demonstrated that critical articulators can be segmented efficiently without task-specific fine-tuning, using global nonlinear filtering and temporal propagation. Their work shows that prompt-based segmentation models can generalize well across speech imaging tasks.

However, their study primarily focuses on segmentation masks as the final output. In contrast, this project treats segmentation as an intermediate step toward extracting interpretable articulatory variables. The proposed work extends beyond mask generation by analysing temporal consistency, articulatory feature trajectories and inter-speaker variability. This addresses an important gap between segmentation performance and speech-science-driven quantitative analysis[6].

Despite these advancements a key limitation remains:

Most existing approaches focus primarily on segmentation accuracy, assuming that accurate masks directly translate to meaningful speech analysis. However, this is not always the case. Errors in segmentation can propagate into articulatory measurements, and prior datasets such as USC-TIMIT have been reported to contain inconsistencies in contour annotations. As a result, there is a lack of reliable pipelines that convert segmentation outputs into stable and

interpretable articulatory variables. This project addresses this gap by explicitly linking segmentation with feature extraction, temporal analysis, and validation of articulatory consistency.

3. Proposed Methodology

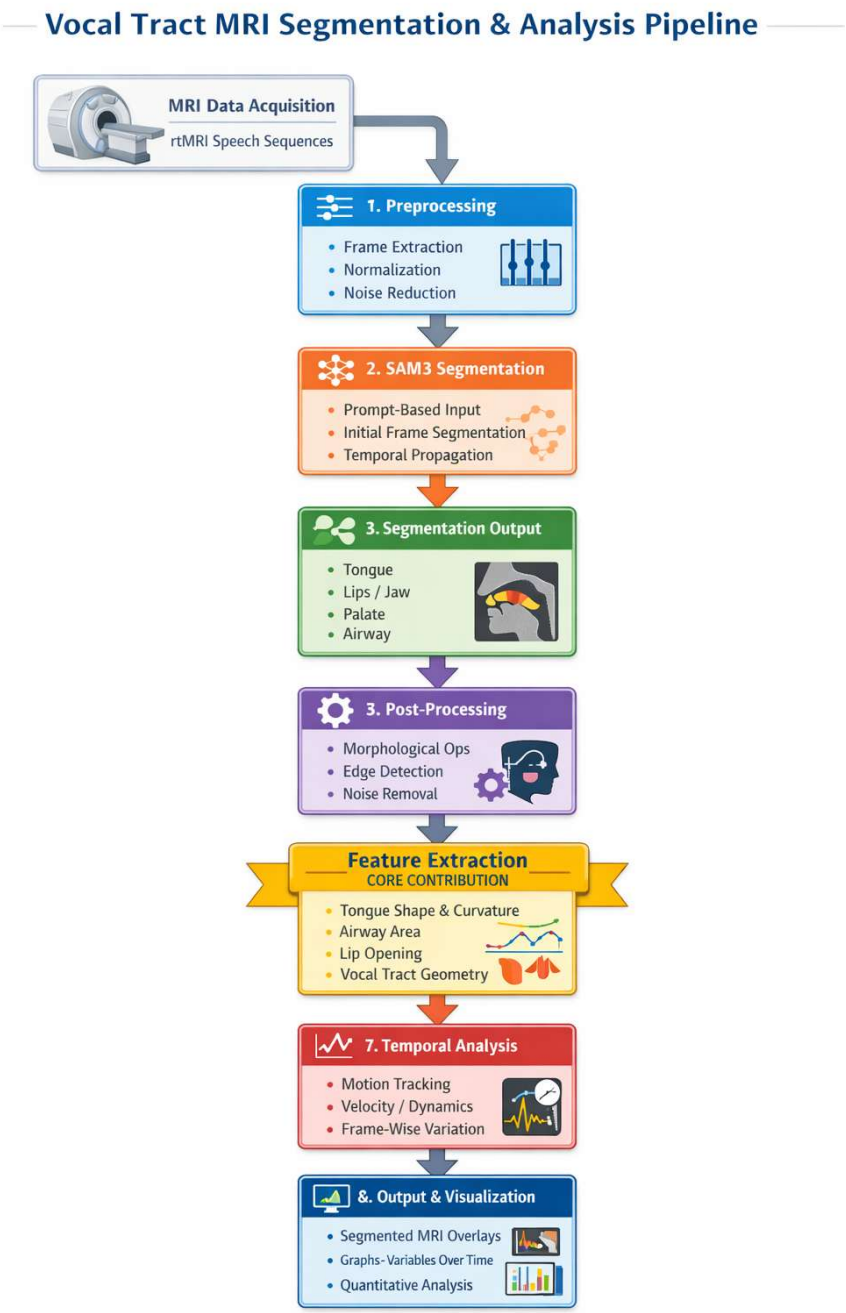


Figure 1. Methodology for project

The proposed method extends SAM-based segmentation by converting segmentation masks into **meaningful articulatory variables and temporal speech patterns**.

3.1 Data Sources

- **USC-TIMIT Dataset** (primary dataset with multiple speakers, used to evaluate generalization and inter-speaker variability) [5].
  Link:https://sail.usc.edu/span/rtmri_ime/
- Optional USC Long Single-Speaker (LSS) dataset (used for controlled analysis of temporal consistency without inter-speaker variation).

USC-TIMIT enables evaluation across multiple speakers, while LSS provides a controlled single-speaker setting to analyse temporal stability of extracted articulatory features.

3.2 Segmentation Approach

- **SAM3 (Segment Anything Model)** for prompt-based segmentation of vocal tract structures [4].
- Segmentation of key articulators:
  - Tongue
  - Lips (upper & lower)
  - Jaw
  - Hard and soft palate
  - Airway
- Temporal propagation across frames for consistent segmentation in MRI sequences.

The proposed method builds upon recent SAM 2.0-based approaches (Hsu et al., 2026) but extends them by incorporating articulatory feature extraction and temporal analysis rather than focusing solely on segmentation masks[6].

3.3 Feature (Variable) Extraction

- Extraction of quantitative articulatory variables from segmentation masks:
  - LA (Lip Aperture): distance between upper and lower lips
  - LP (Lip Protrusion): forward displacement of the lips
  - TT (Tongue Tip position): position of the tongue tip relative to alveolar ridge
  - TB (Tongue Body position): position of the tongue body relative to hard palate
  - TD (Tongue Dorsum position): posterior tongue configuration
  - VL (Velum position**):** velic opening/closure
  - Airway area: midsagittal airspace derived from segmentation masks
  - Vocal tract geometry: derived shape and constriction patterns
- Contour detection and geometric analysis using image processing techniques.

3.4 Tools and Environment

- **Google Colab** for development and GPU-based experimentation.
- **PyTorch** for implementing SAM3 and model-related operations.
- **OpenCV & NumPy** for image processing and feature extraction.
- **Matplotlib** for visualization of articulatory variables over time.

3.5 Evaluation Metrics

Segmentation Performance

- Dice Coefficient
- Intersection over Union (IoU)

These metrics will be computed against available ground truth annotations or reference contour data and compared with prior segmentation approaches including U-Net and SAM-based methods.

Articulatory Feature Evaluation

- Temporal consistency of features (smooth trajectories over frames)
- Robustness to segmentation noise
- Physiological plausibility of articulatory motion

Qualitative Analysis

- Visual inspection of segmentation masks
- Overlay of articulatory trajectories
- Verification of expected speech patterns (e.g., lip aperture changes for vowels)

This multi-level evaluation is necessary because high segmentation accuracy does not always guarantee reliable articulatory feature extraction.

3.6 Inter-Speaker Variability

A key challenge in vocal tract MRI analysis is inter-speaker variability due to anatomical differences and speech patterns. Since USC-TIMIT contains multiple speakers this project will evaluate the robustness of segmentation and feature extraction across different individuals. To address variability the model will incorporate normalization techniques consistent prompt strategies and speaker-wise evaluation. Where possible single-speaker data will be used to isolate temporal performance from anatomical variation.

4. Expected Outcomes

- Development of an **automated segmentation system** capable of accurately identifying key vocal tract structures (tongue, lips, palate, airway) from real-time MRI sequences using SAM3.

- Extraction of **quantitative articulatory variables** articulatory variables such as LA, TT, TB, VL and airway geometry.
- Ability to capture and visualize the **temporal dynamics of speech production**, showing how articulators move and interact across frames.
- Generation of **visual outputs** such as segmented MRI overlays and graphs of articulatory variables over time for analysis and interpretation.
- Achieving segmentation performance with **Dice scores in the range of 0.85–0.90 for major articulators** demonstrating the effectiveness of the SAM3-based approach.

5. Timeline (March – November)

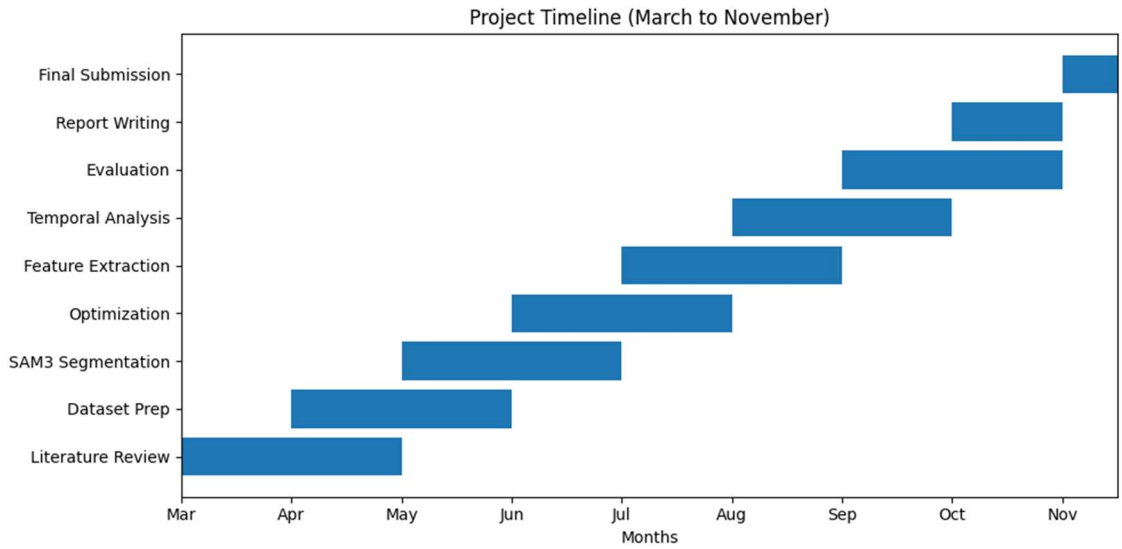


Figure 2: Project Timeline (March to November)

References

[1] M. Ruthven, M. E. Miquel, and A. P. King, “Deep-learning-based segmentation of the vocal tract and articulators in real-time magnetic resonance images of speech,” *Computer Methods and Programs in Biomedicine*, vol. 198, no. 105814, p. 105814, Jan. 2021, doi: https://doi.org/10.1016/j.cmpb.2020.105814.

[2] E. Bresch and S. Narayanan, “Region Segmentation in the Frequency Domain Applied to Upper Airway Real-Time Magnetic Resonance Images,” *IEEE Transactions on Medical Imaging*, vol. 28, no. 3, pp. 323–338, Mar. 2009, doi: https://doi.org/10.1109/tmi.2008.928920.

[3] S. Silva and A. Teixeira, “Unsupervised segmentation of the vocal tract from real-time MRI sequences,” *Computer Speech & Language*, vol. 33, no. 1, pp. 25–46, Sep. 2015, doi: https://doi.org/10.1016/j.csl.2014.12.003.

[4] facebookresearch, “GitHub - facebookresearch/sam3: The repository provides code for running inference and finetuning with the Meta Segment Anything Model 3 (SAM 3), links for downloading the trained model checkpoints, and example notebooks that show how to use the model.,” *GitHub*, Nov. 2025. https://github.com/facebookresearch/sam3

[5] U. of S. C. SAIL LAB, “SAIL: MRI-TIMIT,” *Usc.edu*, 2026. https://sail.usc.edu/span/mri-timit/ (accessed Mar. 19, 2026).

[6] H. Hsu, K. Ng, A. Qureshi, D. Byrd, and Khalil Iskarous, “Automation of real-time vocal tract image segmentation with SAM 2.0 and morphological operation implementation,” *JASA Express Letters*, vol. 6, no. 3, Mar. 2026, doi: https://doi.org/10.1121/10.0042820.

Appendix

- **Repository structure** is already prepared with requirements.txt and placeholder notebook.
- **Future extensions** may include additional MRI datasets to improve generalization and variability

Acknowledgment

I acknowledge the use of digital tools to support the development of this assignment. **ChatGPT (OpenAI, Version: 5, chat.openai.com)** was used to assist in summarizing theoretical concepts, enhancing the clarity of explanations, and refining the overall structure of the content. No text was directly copied, and all AI-generated content was carefully reviewed and edited.

Grammarly was used to check grammar, spelling, and sentence structure, improving the overall readability and correctness of the writing.

Microsoft Power Point was used to create flow diagrams and visual elements that support and complement the written analysis.

The use of these tools complies with the university's policy on AI, and all ideas, interpretations, and final content are entirely my own.

PROJECT PROPOSAL

Title: Segmentation of Vocal Tract using Real-Time MRI Data

Proposer: Tushar Manish Khupte (SID: 520330504)

Supervisor: Dr. Craig Jin

Background: Real-time MRI enables direct visualization of speech articulators such as the tongue, lips, jaw, palate, and airway. However, analysing such data is challenging due to noise, variability and continuous motion. Recent models like SAM 2.0 improve segmentation but lack deeper speech analysis.

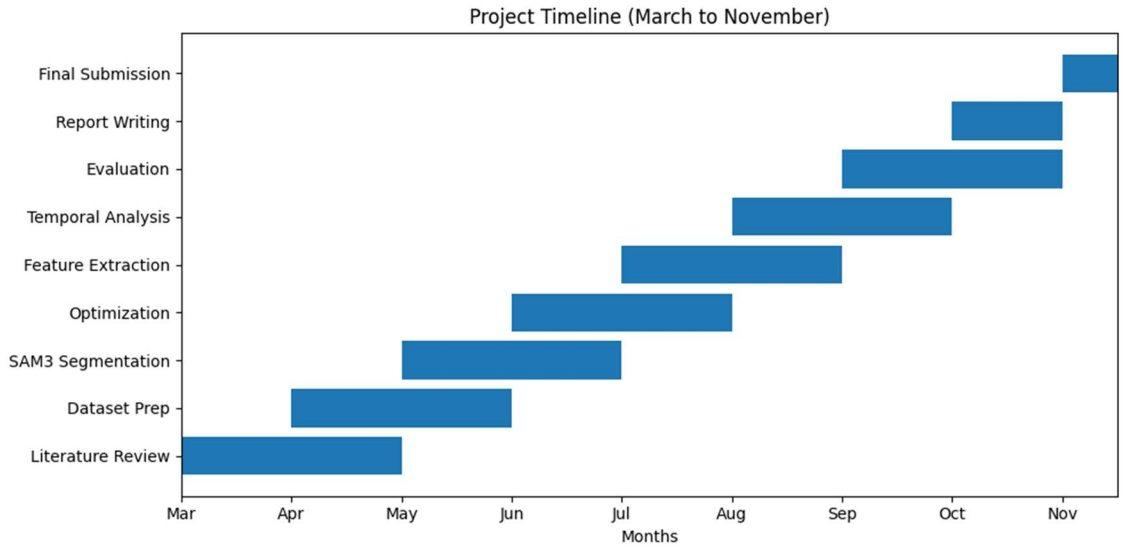
Problem Statement: Existing methods focus only on segmentation and do not extract meaningful articulatory features such as tongue shape, airway area or vocal tract geometry over time.

Methodologies: USC-TIMIT dataset; preprocessing (frame extraction, normalization); SAM3-based segmentation; post-processing; feature extraction (tongue shape, airway area, lip opening); temporal analysis; evaluation using Dice and IoU.

Schedule: Mar–Apr: Literature review & data prep | May–Jun: Segmentation | Jul: Optimization | Aug: Feature extraction | Sep: Temporal analysis | Oct: Evaluation | Nov: Final report

Approval:

Date _____

Supervisor Name _____

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